

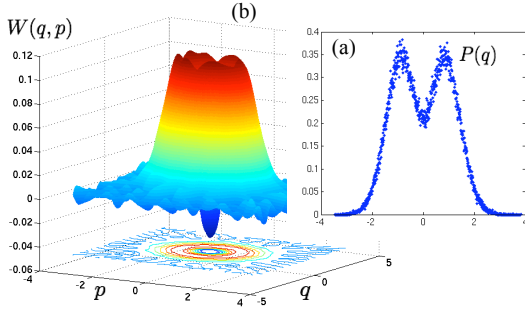


FIG. 2. (a) Raw measured quadrature distribution from the experiment without any correction. (b) Related Wigner function based on the output of the SDP.

certified lower bound to the non-classicality of the state. We will show that the most classical state consistent with those data has a negative Wigner function (Fig. 2.a) which is close to the one reconstructed with a maximum likelihood method in Ref. [9]. This strengthens the finding of a negative Wigner function, as we hence directly certify it as a worst case bound including error bars. Data correspond to phase randomized measurements meaning that we can use the same probability distribution for every phase space direction, the phase being not available in this experiment. Since our non-classicality measure is convex, averaging over phase space directions is an operation which can only decrease the negativity of the state. This means that a lower bound to the non-classicality of the phase randomized state will be valid for the original state.

In order to apply our algorithm we use the measured data to constrain all the points of the characteristic function on a 37×37 lattice. Error bars are estimated using Eq. (7) with $n = 5$ standard deviations. This means that the probability

that all the points of the lattice lie inside the error bars is larger than 99.9%. The lower bound for the non-classicality coming out from the SDP (200 random test vectors have been used) is larger than zero $x' \simeq 0.0028$, meaning that the Wigner function of the state cannot be a positive probability distribution. Indeed, the Wigner function reconstructed from the optimal solution of the SDP (Fig. 2.b) is clearly negative even if we asked for the most positive one consistent with measured data.

Extensions of this approach. – Needless to say, this approach can be extended in several ways. Indeed, the method can readily be generalized to produce lower bounds for *entanglement measures* [20] in the multi-mode setting. Also, this idea can be applied to the situation when not slices are measured, but *points in phase space*, such as when using a detector-atom that is simultaneously coupled to a cantilever [16]. It would also constitute an interesting perspective to see how the present ideas can be used to certify deviations from *stabilizer states* for spin systems, being those states with a positive discrete Wigner function [21].

Summary. – We have introduced a method to directly measure the non-classicality of quantum mechanical modes, requiring less information than tomographic knowledge, but at the same time in a certified fashion. In this way, these ideas are further advocating the paradigm of “learning much from little”—getting much certified information from few measurements—complementing methods of *witnessing entanglement* [20, 22], recent ideas of *compressed sensing* [23] or *matrix-product based* [24] approaches to quantum state tomography, *detector tomography* [25], or the direct estimation of *Markovianity* [26] of a continuous process. It is the hope that this paper further stimulates work in the context of this paradigm.

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I. SUPPLEMENTARY INFORMATION (EPAPS)

In this appendix, we add a discussion on measurement errors not essential for the main text. We assume data to be organized in histograms made of bins of width $2h$ centered at points $s_j = 2hj$. The normalized height of the j -th bin is

$$P_k^j = \int_{s_j-h}^{s_j+h} P_k(s) ds + w_j, \quad (17)$$

where $\{w_i\}$ are random variables with zero average associated to the statistical error. In a standard homodyne detection, $\{w_j\}$ have Poissonian variances equal to $\langle w_j w_l \rangle = \delta_{j,l} \int_{s_j-h}^{s_j+h} P_k(s) ds / N$, where N is the total number of measurements. The simplest approximation for the value of the characteristic function defined in Eq. (6) is

$$\chi^a(\omega \sigma u_k) = \sum_j P_k^j e^{i\omega s_j}. \quad (18)$$

The error will be the sum of two terms

$$\chi(\omega \sigma u_k) - \chi^a(\omega \sigma u_k) = \Delta + \tilde{w}. \quad (19)$$

The first term is the error due to the finite resolution of the histogram, which is bounded by $|\Delta| \leq h|\omega|$, while $\tilde{w} = \sum_j w_j e^{i\omega s_j}$ is the random statistical error. This error could be estimated and possibly reduced by using *bootstrapping* or *maximum likelihood* algorithms. For our purposes, we simply calculate the standard deviation of the random variable $\tilde{w}$, which is given by $(\langle |\tilde{w}|^2 \rangle)^{1/2} = 1/\sqrt{N}$. Now, for the central limit theorem $\tilde{w}$ can be approximated by a Gaussian distribution, meaning that the probability that the following tube constraint is violated is exponentially suppressed in the number of considered standard deviations n [14],

$$|\chi(\omega \sigma u_k) - \chi^a(\omega \sigma u_k)| \leq h|\omega| + n/\sqrt{N}. \quad (20)$$